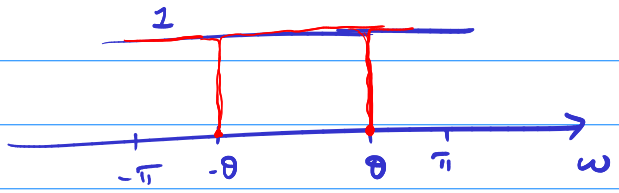
Magnitude Response of LTI Systems

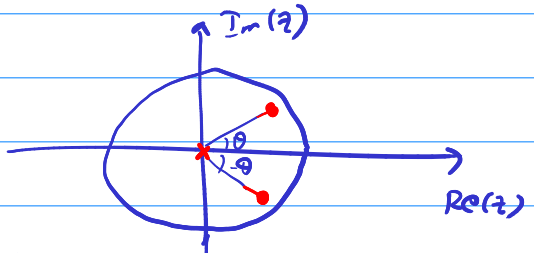
Notch Filter

Want to eliminate one single frequency

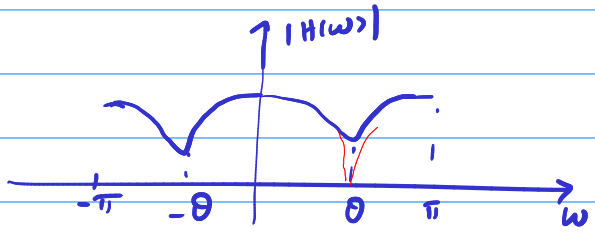
retain all the other frequencies.

$$H(\omega) = \begin{cases} 0 & ; \quad \omega = \pm\theta \\ 1 & ; \quad \omega \neq \pm\theta \end{cases}$$



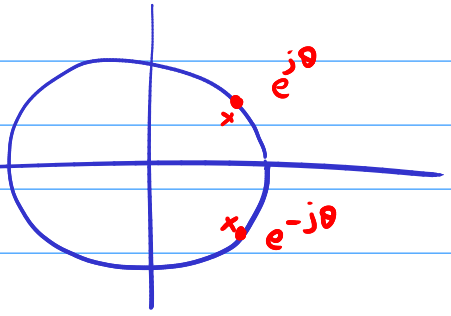


$$H(z) = \left(1 - re^{+j\theta} z^{-1}\right)\left(1 - re^{-j\theta} z^{-1}\right)$$

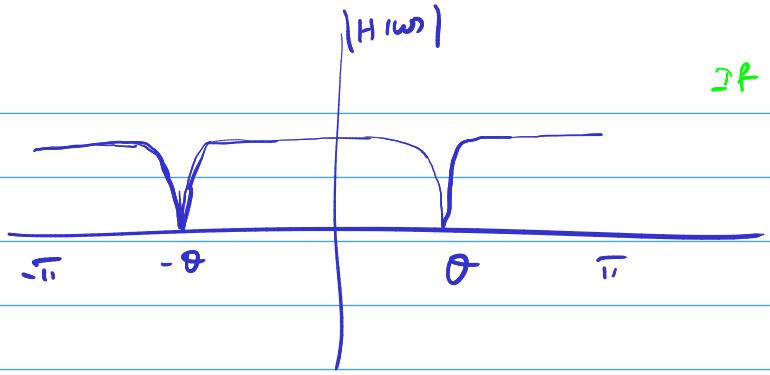


Improved Notch filter

$$H(z) = \frac{\left(1 - e^{j\theta} z^{-1}\right)\left(1 - e^{-j\theta} z^{-1}\right)}{\left(1 - re^{j\theta} z^{-1}\right)\left(1 - re^{-j\theta} z^{-1}\right)}$$



$r < 1$

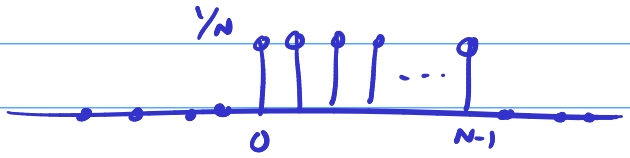


If r is close to 1 then response gets close to ideal Notch filter



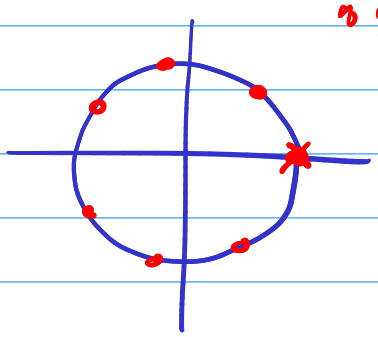
(Causal) Moving Average Filter

$$h(n) = \begin{cases} 1/N ; & 0 \le n \le N-1 \\ 0 ; & \text{else} \end{cases}$$



Pole zero Cancellation at $z=1$

Zeros are N^{th} roots of unity.

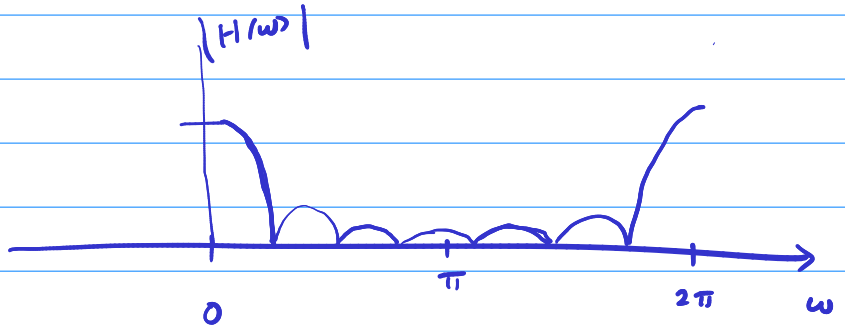


$$H(z) = \sum_{n=0}^{N-1} \frac{1}{N} z^{-n}$$

$$= \frac{1}{N} \frac{1 - z^{-N}}{1 - z^{-1}}$$

$$H(\omega) = \frac{1}{N} \frac{1 - e^{-j\omega N}}{1 - e^{-j\omega}}$$

$$|H(\omega)| = \frac{1}{N} \frac{|1 - e^{-j\omega N}|}{|1 - e^{-j\omega}|} = \frac{1}{N} \frac{|\sin(\omega N/2)|}{|\sin(\omega/2)|}$$



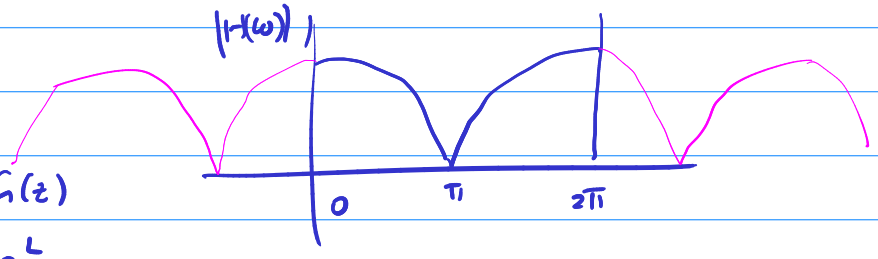
Comb Filter

→ Sampling rate conversion.

Consider a first order FIR filter

$$H(z) = 1 + z^{-1}$$ LPF

$$h(n) = \{1, 1\}$$



we obtain new transfer function $G(z)$

Replace z with z^L in $H(z)$

L is a positive integer.

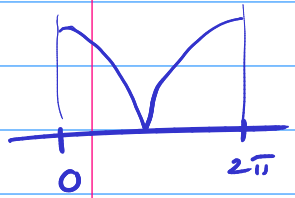
$$G(z) = H(z^L) = 1 + z^{-L}$$

$$g(n) = \{1, \underbrace{0, 0, \ldots, 0}_{(L-1) \text{ zeros}}, 1\}$$

$x(t) \rightarrow$ CT sig^{n}

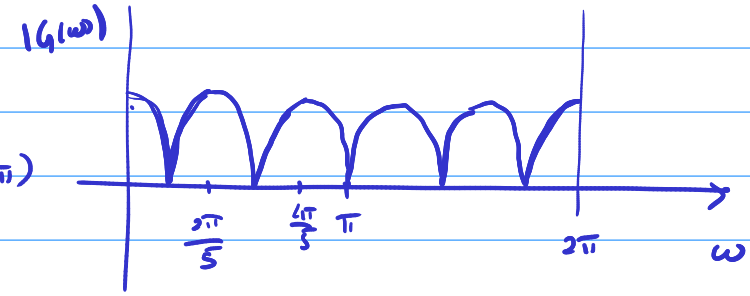
$x(at) \rightarrow$ time scaled sig^{n}

$$H(\omega) = 1 + e^{-j\omega}$$



$$G(\omega) = 1 + e^{-j\omega L} = H(\omega L)$$

$L = 5$



$$G(\omega = 0) = H(\omega = 0)$$

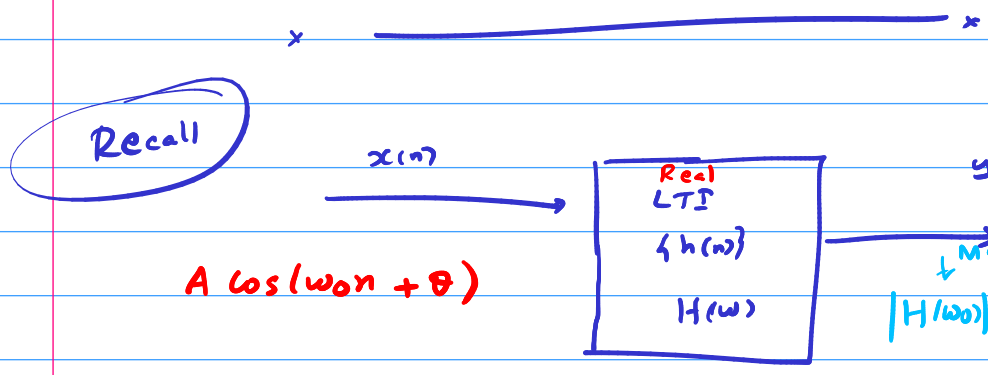
$$G\left(\omega = \frac{\pi}{L}\right) = H\left(\frac{\pi}{L} \cdot L\right) = H(\pi)$$

$$G\left(\frac{2\pi}{L}\right) = H(2\pi)$$

We can do comb filter with HPF

$$H(z) = 1 - z^{-1}$$

$$G(z) = 1 - z^{-L}$$



$$H(\omega) = |H(\omega)| e^{j \angle H(\omega)}$$

$\angle H(\omega) \Rightarrow$ Phase response as a function of ω.